



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

25-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,C	A,B,D	A,C	A,C	A,C,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	2.00	4.00	4.00	169.40 to 169.60	20.00	3.00	6.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	C	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	B,C	A,B	B,C,D	A,B,C,D	A,B,C,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	0.33 or 0.34	99.98	1.25	6.00	4.00	0.375	1.92 to 1.93

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,D	B,C	C,D	A,B,C	A,C,D	A,C		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	100.00	4.00	25.00	5.20	2.00	8.00	3.00	3.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

Resultant amplitude

$$= \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \Delta\phi}$$

$$a^2 = 2a^2 + 2a^2 \cos(\phi_1 - \phi_2)$$

$$\Rightarrow \cos(\phi_1 - \phi_2) = -\frac{1}{2}$$

$$\phi_1 - \phi_2 = 120^\circ \text{ or } \frac{2\pi}{3} \text{ rad}$$

OR

For superposition of two oscillations

$$a \sin(\omega t + \phi) = a \sin(\omega t + \phi_1) + a \sin(\omega t + \phi_2)$$

$$\Rightarrow \sin \omega t \cos \phi + \cos \omega t \sin \phi$$

$$= \sin \omega t \cos \phi_1 + \cos \omega t \sin \phi_1 + \sin \omega t \cos \phi_2 + \cos \omega t \sin \phi_2$$

$$\Rightarrow \cos \phi = \cos \phi_1 + \cos \phi_2$$

$$\& \sin \phi = \sin \phi_1 + \sin \phi_2$$

$$\Rightarrow \cos^2 \phi + \sin^2 \phi = 1 = \cos^2 \phi_1 + \cos^2 \phi_2 +$$

$$2\cos \phi_1 \cos \phi_2 + \sin^2 \phi_1 + \sin^2 \phi_2 + 2\sin \phi_1 \sin \phi_2$$

$$\Rightarrow \cos(\phi_1 - \phi_2) = -\frac{1}{2}$$

$$\Rightarrow \phi_1 - \phi_2 = 120^\circ \text{ or } \frac{2\pi}{3} \text{ rad}$$

2. **Ans (B)**

Field inside conductor is zero. Also field on the surface is perpendicular to the surface for conductor.

3. **Ans (D)**

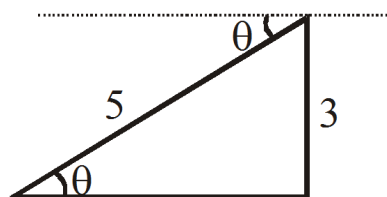
E & B are perpendicular so (4) is not possible

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C)**

$$V_{\text{Ring1}} = V_{\text{Ring2}}$$



$$\sin \theta = \frac{3}{5}$$

$$\Rightarrow \theta = 37^\circ$$

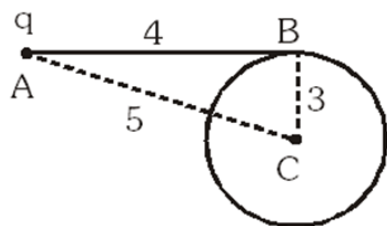
$$V_{\text{Ring1}} \times \cos 37^\circ = 5 \text{ m/sec}$$

$$\Rightarrow V_{\text{Ring1}} = \frac{5}{4} \times 5 = \frac{25}{4} \text{ m/sec}$$

6. **Ans (A,C)**

$$V = \frac{kq}{AC} = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{5}$$

$$= 1.8 \times 10^3 = 1.8 \text{ kV}$$



$$V_B = (V_B)_{\text{due to } q} + (V_B)_i$$

Here $(V_B)_i$ = Potential at B due to induced charge

$$\therefore 1.8 \times 10^3 = \frac{kq}{AB} + (V_B)_i$$

$$\Rightarrow 1.8 \times 10^3 = 2.25 \times 10^3 + (V_B)_i$$

$$\Rightarrow (V_B)_i = -0.45 \text{ kV}$$

7. **Ans (A,B,D)**

$$V_A = \frac{Kq_A}{R} + \frac{Kq_B}{2R} = 2V,$$

$$V_B = \frac{K(q_A + q_B)}{2R} = \frac{3}{2}V$$

$$\Rightarrow \frac{q_A}{q_B} = \frac{1}{2}$$

$$\text{After earthing of B, } q'_A = q_A \text{ \& } V'_B = \frac{K(q'_A + q'_B)}{2R} = 0$$

$$\Rightarrow \left| \frac{q'_A}{q_B} \right| = 1 \text{ \& } V'_A - V'_B = \frac{V}{2}$$

8. **Ans (A,C)**

Q value for fusion =

$$(B/A)_{75} \times 75 - [(B/A)_{30} \times 30 + (B/A)_{45} \times 45]$$

$$= 99 \text{ MeV.}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (3.00)**

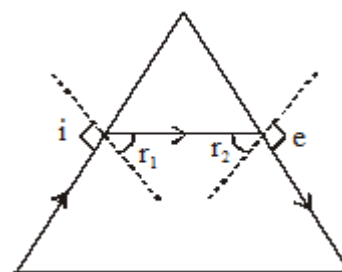
Frame will complete vertical circular motion if it crosses position where line joining O and centre of mass C makes an angle

$$\theta = \tan^{-1} \frac{a}{g} = 37^\circ \text{ with vertical}$$

By work energy theorem

$$\frac{1}{2} (2m\ell^2) \omega^2 = 2mg \frac{\sqrt{3}}{2} \ell (1 + \cos \theta) + 2ma \frac{\sqrt{3}}{2} \ell \sin \theta$$

2. **Ans (2.00)**



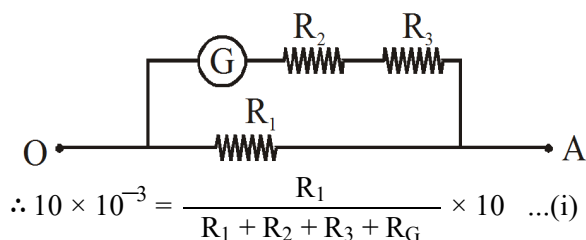
$$\sin i = \sqrt{\mu^2 - 1} \sin r_1 - \cos r_2$$

since light strike grazingly so $i = 90^\circ$ r_1 and $r_2 = 60^\circ$

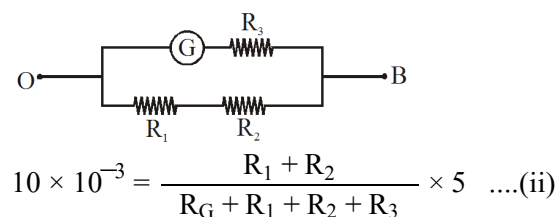
$$\text{so, } \mu = 2$$

3. Ans (4.00)

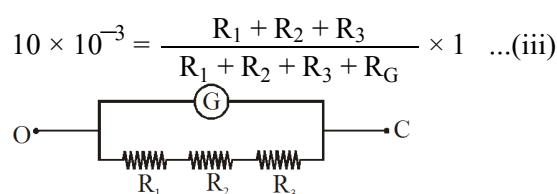
For connection between OA,



For connection between OB,



For connection between OC,

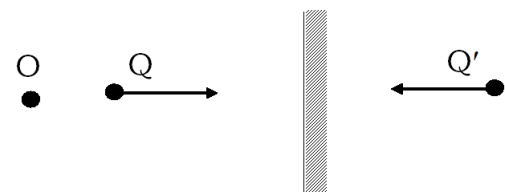


Solving (i), (ii) & (iii)

$$R_1 = \frac{1}{10} \Omega, R_2 = \frac{1}{10} \Omega \text{ \& } R_3 = \frac{4}{5} \Omega$$

5. Ans (169.40 to 169.60)

Let f_1 and f_2 be the frequencies of tuning forks P and Q, Then $|f_1 - f_2| = 7/2$



Apparent frequency for O corresponding to signal directly coming from

$$Q = f_2 \left(\frac{v}{v + v_q} \right)$$

Apparent frequency of the echo = $f_2 \left(\frac{v}{v - v_q} \right)$

$$\therefore \Delta f_2 = f_2 \left[\frac{2v_q v}{v^2 - v_q^2} \right]$$

Since, $\Delta f_2 = 5$ (given), $\therefore f_2 = 166$ Hz. Now, $f_1 = 166 \pm 3.5 = 169.50$ or 162.50 Hz.

When P is filed, its frequency will increase, since it is given that filed P gives greater number of beats with Q. It implies that f_1 must be 169.50 Hz.

6. Ans (20.00)

Velocity of first block before collision.

$$v_1^2 = 1^2 - 2(4) \times 0.08 = 1 - 0.64$$

$$v_1 = 0.6 \text{ m/s}$$

By conservation of momentum, $3 \times 0.6 = 3v_1' + 6v_2'$

also $v_2' - v_1' = v_1$ for elastic collision

It gives $v_2' = 0.4 \text{ m/s}$

$$v_1' = -0.2 \text{ m/s}$$

Now distance moved after collision

$$s_1 = \frac{(0.4)^2}{2 \times 4} \text{ \& } s_2 = \frac{(0.2)^2}{2 \times 4}$$

$$\therefore x = s_1 + s_2 = 0.025 \text{ m} = 2.5 \text{ cm.}$$

7. Ans (3.00)

$$\frac{1}{2} \times \left(\frac{m\ell^2}{3} \right) \left(\frac{V}{\ell} \right)^2 = mg \left(\frac{\ell}{2} \right)$$

$$\Rightarrow V = \sqrt{3g\ell} = 3$$

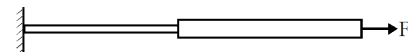
8. Ans (6.00)

$$\frac{F}{A_1} = y \frac{\Delta \ell_1}{2\ell}$$

$$\frac{F}{A_2} = y \frac{\Delta \ell_2}{\ell}$$

$$\Delta \ell_1 + \Delta \ell_2 = \frac{27}{4} \text{ mm}$$

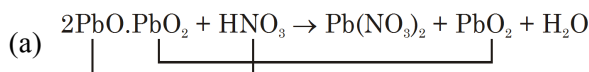
$$\frac{2F\ell}{A_1 y} + \frac{F\ell}{4A_1 y} = \frac{27}{4} \text{ mm}$$



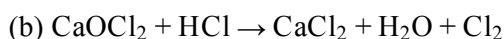
PART-2 : CHEMISTRY

SECTION-I (i)

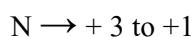
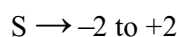
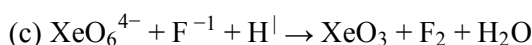
1. Ans (A)



Here, PbO is behaving as base while PbO₂ is filtered out.



Cl is undergoing disproportionation



3. **Ans (C)**

At low pressure,

$$\left(P + \frac{a}{V^2}\right)(V) = RT$$

$$\text{i.e. } PV + \frac{a}{V} = RT$$

$$\text{i.e. } PV^2 - RTV + a = 0$$

$$V = \frac{RT \pm \sqrt{R^2T^2 - 4Pa}}{2P}$$

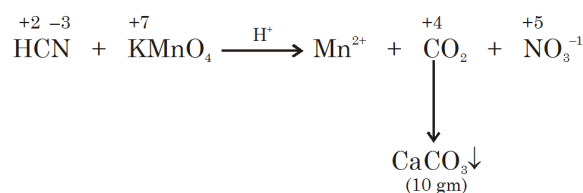
$$\text{as } R^2T^2 = 4Pa$$

$$\therefore V = \frac{RT}{2P}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A,B)**



n-factor of HCN = 10

$$\therefore N = 1 \times 10 \Rightarrow 10N$$

$$\text{Eq. mass} = \frac{M}{n\text{-factor}} = \frac{M}{10} \Rightarrow E < M$$

At eq. point :

$$N_1V_1 = N_2V_2 \Rightarrow 10 \times 100 = 5 \times 5 \times V$$

$$\Rightarrow V = 40\text{ml}$$

8. **Ans (B,C,D)**

Refer theory

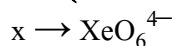
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.00)**



2. **Ans (0.33 or 0.34)**



3. **Ans (99.98)**

$[\text{Ag}^+]$ for pptⁿ of I^-

$$= \frac{k_{sp}(\text{AgI})}{[\text{I}^-]} = \frac{10^{-17}}{0.1} = 10^{-16} \text{M}$$

$[\text{Ag}^+]$ for pptⁿ of Br^-

$$= \frac{k_{sp}(\text{AgBr})}{[\text{Br}^-]} = \frac{10^{-13}}{0.2} = \frac{1}{2} \times 10^{-12} \text{M}$$

$\Rightarrow [\text{I}^-]$ when AgBr starts precipitating

$$= \frac{k_{sp}(\text{AgI})}{[\text{Ag}^+]} = \frac{10^{-17}}{10^{-12}} \times \frac{1}{2} = 2 \times 10^{-5}$$

$\Rightarrow \% [\text{I}^-]$ remaining in solution

$$= 2 \times \frac{10^{-5}}{0.1} \times 100 \text{ for precipitation of AgBr}$$

$$\Rightarrow \% [\text{I}^-] \text{ precipitated} = 100 - 0.02 = 99.98$$

4. **Ans (1.25)**

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 = T, T_2 = \frac{T}{2}$$

$$V_1 = V \text{ and } V = 5.66 V$$

$$T V^{\gamma-1} = \frac{T}{2} (5.66 V)^{\gamma-1};$$

$$2 = (5.66) \gamma^{\gamma-1}$$

$$\gamma = 1.4$$

$$|w| = \left| \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} \right| = \frac{nRT}{2(\gamma - 1)} = \frac{1}{0.8} PV = 1.25 PV$$

5. **Ans (6.00)**

(i), (ii), (iv), (v), (vi), (viii)

7. **Ans (0.375)**

Leucine; Lysine; Threonine are essential amino acid;

$\therefore x = 3$ linkage in phosphodiester is 3 - 5

$$\therefore y = 3; z = 5$$

$$\therefore \text{Ans} = \frac{3}{8} = 0.375$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

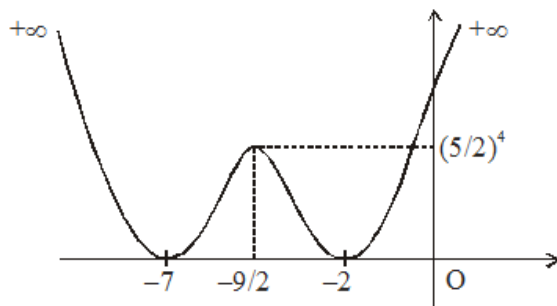
$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \dots \sec(e^{50}x))}{e^{2 \cos x} \frac{(e^{2-2 \cos x}-1)}{2-2 \cos x} \cdot \frac{2-2 \cos x}{x^2} \cdot x^2} (\%) \\ &= \lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x))}{e^{2x^2}} (\%) \\ &= \lim_{x \rightarrow 0} \frac{\ln(1 + \sec(ex) \dots \sec(e^{50}x) - 1)}{e^{2x^2}} \\ & \left[\text{Use } \frac{\ln(1+x)}{x} \text{ limit} \right] \\ &= \lim_{x \rightarrow 0} \frac{\sec(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x) - 1}{e^{2x^2}} \end{aligned}$$

Apply LH rule

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{e \sec(ex) \tan(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x) + \text{so on}}{2e^{2x}} \\ &= \lim_{x \rightarrow 0} \frac{e^2 + e^4 + \dots + e^{100} \rightarrow (\text{G.P.})}{2e^2} \\ &= \lim_{x \rightarrow 0} \frac{e^2((e^2)^{50} - 1)}{2e^2(e^2 - 1)} = \frac{e^{100} - 1}{2(e^2 - 1)} \end{aligned}$$

2. Ans (C)

$$\begin{aligned} \text{let } f(x) &= (x+2)^2(x+7)^2 \\ f'(x) &= 2(x+2)(x+7)^2 + (x+2)^2(x+7) \\ &= 2(x+7)(x+2)(2x+9) \end{aligned}$$



for $a = 0$, given equation has exactly two double roots ($x = -2, x = -7$).

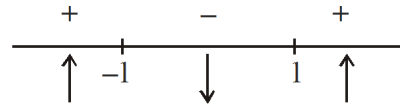
for $a = -\left(\frac{5}{2}\right)^4$, given equation has exactly one double root. $\left(x = -\frac{9}{2}\right)$

[double root $\Rightarrow$ function and the derivative both are zero at that x]

3. Ans (B)

$$f(x) = x^3 - 3x + 1$$

$$f'(x) = 3(x^2 - 1)$$



$$f(-1) = 3, f(0) = 1, f(2) = 3$$

Checking other nearby points.

$$f(-2) = -1, f(1) = -1$$

$$f(-2), f(-1) < 0 \quad (0, 1) \quad (1, 2)$$

$(-2, -1)$ one root

$$x_1 \in (-2, -1), x_2 \in (0, 1), x_3 \in (1, 2)$$

$$[x_1] = -2 \quad [x_2] = 0 \quad [x_3] = 1$$

$$\text{and } x_1 + x_2 + x_3 = 0 \quad (\text{sum of roots})$$

$$\therefore \{x_1\} + \{x_2\} + \{x_3\}$$

$$= x - [x_1] + x - [x_2] + x - [x_3]$$

$$= 0 - (-2 + 0 + 1) = 1$$

4. Ans (B)

$$P_1 + \lambda P_2 = 0$$

$$(x + 2y + z - 10) + \lambda(3x + y - z - 5) = 0$$

passing through $(0, 0, 0)$

$$\Rightarrow -10 + \lambda(-5) = 0 \Rightarrow \lambda = -2$$

$$\therefore \text{plane is } (x + 2y + z - 10) - 2(3x + y - z - 5) = 0$$

$$\Rightarrow 5x - 3z = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,D)

$$\text{Let } \int_0^{\pi/4} \cos t \cdot f(t) dt = A$$

$$\therefore f(x) = \sin^2 x \cos^{-5} x + \mu A \sin^2 x \cos^{-5} x = (1 + \mu A)$$

$$\sin^2 x \cos^{-5} x$$

$$A = \int_0^{\pi/4} \cos t (1 + \mu A) \sin^2 t \cos^{-5} t dt$$

$$A = (1 + \mu A) \int_0^{\pi/4} \frac{\sin^2 t}{\cos^4 t} dt$$

$$= (1 + \mu A) \int_0^{\pi/4} \sec^2 t \tan^2 t dt$$

$$[\text{put } \tan t = z \sec^2 t dt = dz]$$

$$A = (1 + \mu A) \left(\frac{\tan^3 t}{3} \right)_0^{\pi/4} = \frac{1 + \mu A}{3}$$

$$\Rightarrow 1 + \mu A = 3A \text{ and } (3 - \mu)A = 1 \Rightarrow A = \frac{1}{3 - \mu}$$

$$\therefore f(x) = \frac{3}{3 - \mu} \sin^2 x \cos^{-5} x$$

$$(A)(D) \text{ period} = 2\pi$$

6. Ans (A,C,D)

$$f(f(x)) = -x + 1$$

$$\text{replace } x \rightarrow f(x)$$

$$f(f(f(x))) = -f(x) + 1$$

$$f(1 - x) = -f(x) + 1$$

$$\Rightarrow f(x) + f(1 - x) = 1 \quad \dots(i)$$

$$\therefore f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) = 1 \quad (\text{B correct})$$

$$\text{And, } I = \int_0^1 f(x) dx = \int_0^1 f(1 - x) dx$$

$$2I = \int_0^1 f(x) + f(1 - x) dx$$

$$\Rightarrow I = \frac{1}{2} \quad (\text{A correct})$$

Twice differentiating (i), we get

$$f''(x) + f''(1 - x) = 0$$

$$\Rightarrow f''\left(\frac{1}{2}\right) = 0 = f''\left(\frac{4}{5}\right)$$

$$\text{By Roll's theorem. } f'''(x) = 0 \Rightarrow x \in \left(\frac{1}{2}, \frac{4}{5}\right)$$

$$\text{Also, } f'(x) = f'(1 - x) \Rightarrow f'\left(\frac{1}{5}\right) = f'\left(\frac{4}{5}\right)$$

$$\text{so by Roll's theorem } f''(x) = 0 \Rightarrow x \in \left(\frac{1}{5}, \frac{4}{5}\right)$$

7. Ans (C,D)

$$\vec{b} = 2\hat{c} + \lambda\hat{a}$$

$$|\vec{b}|^2 = 4 + \lambda^2 + 4\lambda\hat{c} \cdot \hat{a}$$

$$|\vec{b}|^2 = 4 + \lambda^2 + 4\lambda\left(\frac{1}{7}\right) = 16$$

$$\Rightarrow \lambda^2 + \lambda - 12 = 0$$

$$\Rightarrow \lambda = -4, 3$$

8. Ans (A,B,C)

(A) $|Z| = r, \arg z = \theta, z = x + iy$

$|x + iy - 2 - i| r \left| \sin\left(\frac{\pi}{4} - \theta\right) \right|$

$|(x-2) + i(y-1)| = r \left| \sin \frac{\pi}{4} \cos \theta - \cos \frac{\pi}{4} \sin \theta \right|$

$|(x-2) + i(y-1)| = r \left(\frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{2}} \sin \theta \right)$

$\sqrt{(x-2)^2 + (y-1)^2} = \frac{1}{\sqrt{2}} |x-y|$

$x^2 + 4 - 4x + y^2 + 1 - 2y = \frac{1}{2}(x-y)^2$

$\Rightarrow (x+y)^2 = (4x+2y-5)$

(B) $z = \text{constant} \Rightarrow \text{incorrect}$

(C) $z = a\omega, a\omega^2 \Rightarrow \arg = \frac{2\pi}{3}$

9. Ans (A,C,D)

$\frac{1}{y^3} \frac{dy}{dx} - \frac{1}{y^2} \cdot \frac{1}{x} = -(1 + \cos x)$

let $\frac{-1}{y^2} = v$

$\frac{2}{y^3} \frac{dy}{dx} = \frac{dv}{dx}$

$\frac{1}{2} \frac{dv}{dx} + \frac{v}{x} = -(1 + \cos x)$

$\Rightarrow \frac{dv}{dx} + \frac{2}{x}v = -2(1 + \cos x)$

u. $v^2 = -2 \int x^2(1 + \cos x)dx$

$= \frac{x^3}{3} + v^2 \sin x + 2x \cos x - 2 \sin x + k$

10. Ans (A,C)

$$I = 2 \int_0^{\pi/2} \sin x (1 - \cos^2 x) e^{-\sin^2 x} dx$$

$$2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \int_0^{\pi/2} \underbrace{(\cos x)}_I \underbrace{(-\sin 2x)}_{II} e^{-\sin^2 x} dx \quad (\text{IBP})$$

$$2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \left(\cos x \cdot e^{-\sin^2 x} \right)_0^{\pi/2} + \int_0^{\pi/2} \sin x \cdot e^{-\sin^2 x} dx$$

$$= 3 \int_0^{\pi/2} \sin x \cdot e^{-\sin^2 x} dx - 1$$

put $-\sin^2 x = t \Rightarrow \cos^2 x = 1+t$

$-2 \sin x \cos x dx = dt$

$\sin x dx = \frac{-dt}{2 \cos x} = \frac{-dt}{2\sqrt{1+t}}$

$I = \frac{-3}{2} \int_0^{-1} \frac{e^t dt}{\sqrt{1+t}} - 1 \quad (\text{put } 1+t = z)$

$= \frac{-3}{2} \int_1^0 \frac{e^{z-1}}{\sqrt{z}} dz$

$= \frac{3}{2e} \int_0^1 e^z \frac{1}{\sqrt{z}} dz - 1 \quad (\text{IBP})$

$= \frac{3}{2e} \left([2\sqrt{z} e^z]_0^1 - 2 \int_0^1 e^z \sqrt{z} dz \right) - 1$

$= 2e$

$= 3 - \frac{3}{e} \int_0^1 e^z \sqrt{z} dz - 1$

$= (2) - (3) e \int_0^1 e^z \sqrt{z} dz \Rightarrow \alpha = 2, \beta = 3$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (100.00)

$[\{x\}] = 0$

$f(x) = (3-x^7)^{\frac{1}{7}}$

2. Ans (4.00)

${}^9C_3 \times {}^3C_2 \times \frac{5!}{2!2!1!}$

3. Ans (25.00)

$$(\sin x + \cos x)(\tan^2 x - (1 + \sqrt{3})\tan x + 3 + \sqrt{3}) = 3|\sin x + \cos x|$$

Case I :

$$\sin x + \cos x \geq 0$$

$$(s + c)(t^2 - (1 + \sqrt{3})t + \sqrt{3}) = 0$$

$$\Rightarrow t = -1, 1, \sqrt{3}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{7\pi}{4}$$

Case II :

$$s + c < 0$$

$$(s + c)(t^2 - (1 + \sqrt{3})t + 6 + \sqrt{3}) = 0$$

$$\Rightarrow t = -1 \quad \text{No solution}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\Rightarrow \text{sum of solutions is } \frac{37\pi}{12}$$

$$|p - q| = 25$$

4. Ans (5.20)

$$t_1 = \frac{\sin x}{\cos 3x}, t_2 = \frac{\sin 3x}{\cos 9x}, t_3 = \frac{\sin 9x}{\cos 27x}$$

$$t_1 = \frac{\sin x \cdot 2 \cos x}{\cos 3x \cdot 2 \cos x} = \frac{\sin 2x}{2 \cos x \cos 3x} = \frac{\sin(3x - x)}{2 \cos x \cos 3x}$$

$$= \frac{1}{2}(\tan 3x - \tan x)$$

$$\text{Similarly, } t_2 = \frac{1}{2}(\tan 9x - \tan 3x)$$

$$t_3 = \frac{1}{2}(\tan 27x - \tan 9x)$$

$$t_1 + t_2 + t_3 = 0 \Rightarrow \frac{1}{2}(\tan 27x - \tan x) = 0$$

$$\Rightarrow \tan 27x = \tan x$$

$$\Rightarrow 27x = n\pi + x$$

$$\Rightarrow 26x = n\pi \Rightarrow x = \frac{n\pi}{26}$$

$$n = 1 \Rightarrow x = \frac{\pi}{26} \therefore k = 26$$

5. Ans (2.00)

$$x + y + z = \theta \Rightarrow \cos x + \cos y + \cos z = 2 \cos \theta$$

$$\& \sin x + \sin y + \sin z = 2 \sin \theta$$

$$\& \cos(x + y) + \cos(y + z) + \cos(z + x) = ?$$

$$= \cos(\theta - z) + \cos(\theta - x) + \cos(\theta - y)$$

$$= \cos \theta \cos z + \sin \theta \sin z + \cos \theta \cos x + \sin \theta \sin x$$

$$+ \cos \theta \cos y + \sin \theta \sin y$$

$$= \cos \theta (\cos x + \cos y + \cos z) + \sin \theta (\sin x + \sin y + \sin z)$$

$$= 2 \cos^2 \theta + 2 \sin^2 \theta = 2$$

6. Ans (8.00)

$$\text{Replace } x \rightarrow \frac{2}{x}$$

$$\left(\frac{8}{x^2} + \frac{6}{x} + 4 \right)^{10} = \sum_{r=0}^{20} a_r \left(\frac{2}{x} \right)^r$$

$$\Rightarrow \frac{2^{10}(2x^2 + 3x + 4)^{10}}{x^{20}} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{-r}$$

$$\Rightarrow 2^{10}(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$$\Rightarrow 2^{10} \sum_{r=0}^{20} a_r \cdot x^r = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$$\therefore \text{coefficient of } x^7$$

$$2^{10} a_7 = a_{13} 2^{13}$$

$$\Rightarrow \frac{a_7}{a_{13}} = 2^3 = 8$$

7. Ans (3.00)

$$\text{Let } f(x) = \tan^{-1} x, x \in [a, b]$$

$$\therefore \text{Using LMVT, we get } \frac{\tan^{-1} b - \tan^{-1} a}{b - a} = \frac{1}{1 + c^2}$$

$$\text{, where } 0 < a < c < b < \sqrt{3}$$

$$\text{So, } 1 < \left(\frac{b - a}{\tan^{-1} b - \tan^{-1} a} \right) < 4$$

$$\text{As, } \frac{1}{4} < \frac{1}{1 + c^2} < 1$$

$$\Rightarrow \text{The greatest possible integral value is 3.}$$

8. Ans (3.00)

$$\int_0^{\pi/2} \frac{dx}{\cos^2 x (1 + \sqrt{\tan x})^4} = \int_0^{\pi/2} \frac{\sec^2 x}{(1 + \sqrt{\tan x})^4} dx$$

$$\text{Put } \tan x = t^2 \Rightarrow \sec^2 x dx = 2t dt$$

$$\int_0^{\infty} \frac{2t}{(1 + t)^4} dt = 2 \int_0^{\infty} \left(\frac{1}{(1 + t)^3} - \frac{1}{(1 + t)^4} \right) dt$$

$$= 2 \left(\frac{-1}{2(1 + t)^2} + \frac{1}{3(1 + t)^3} \right)_{\infty}^0$$

$$= \left(\frac{-1}{2(1 + t)^2} + \frac{1}{3(1 + t)^3} \right)_0^{\infty}$$

$$= 2 \left(0 - \left(\frac{-1}{2} + \frac{1}{3} \right) \right) = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$